Huawei Access Node JAY-ACC-60 - Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) Troubleshooting and Resolution SOP

Node: JAY-ACC-60 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_5529

Description

The Huawei Access Node JAY-ACC-60, situated at the Jayanagar Metro Site, is an integral component of the optical transport domain. This node, being an Access node, is responsible for establishing connections between the customer premises and the metropolitan network. The primary failure domain of interest in this SOP is the Optical Transport Failure, encompassing issues such as SFP/XFP degradation, fiber cuts, and Rx/Tx power loss. These failures can occur due to various reasons, including hardware degradation, environmental factors, or human error. In the context of Huawei equipment, the Optical Transport Network (OTN) protocol plays a crucial role in ensuring reliable data transmission. However, when SFP/XFP degradation or fiber cuts occur, the OTN protocol can be compromised, leading to packet loss, corruption, or even complete network outages. The Rx/Tx power loss issue can also be a result of SFP/XFP degradation or improper fiber connection. Therefore, it is essential to identify and rectify these issues promptly to prevent network downtime and ensure uninterrupted service delivery.

Pre-requisite Checks

- Verify the site location and ensure remote hands are dispatched to the Jayanagar Metro Site for on-site support.

- Check the redundant link status to ensure that it is operational and ready to take over in case of a failure.

- Gather all relevant information about the failure, including the alarm code (ALARM_5529) and severity (Minor).

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display optics interface brief, Expected Output: Interface Tx
Power(Rx) Rx Power(Tx) Status
------------------------------------------------- GE0/0/0 - - Not
Present GE0/0/1 - - Not Present , Description: This command is used to
display the optical interface information, including Tx and Rx power
levels. The output indicates if the interface is not present, which
can be a sign of a fiber cut or SFP/XFP degradation.

Command: display optics interface detail GE0/0/0, Expected Output:
Optics Interface Detail for GE0/0/0
------------------------------------------------- Current Status:
Not Present Last Changed: Never SFP Type: GE-SFP-1000BASE-LX/LH SFP
Serial Number: 1234567890 , Description: This command provides
detailed information about the GE0/0/0 interface, including its
current status, SFP type, and serial number. The output indicates that
the interface is not present, which can be a sign of SFP/XFP
degradation or fiber cuts.

Command: display optics interface statistics GE0/0/0, Expected Output:
Optics Interface Statistics for GE0/0/0
------------------------------------------------- Total Rx Errors: 0
Total Tx Errors: 0 Last Clear Time: Never , Description: This command
displays the statistics for the GE0/0/0 interface, including total Rx
and Tx errors. The output indicates that there are no errors, which
can be a sign of proper fiber connection and SFP/XFP functionality.
```

Resolution Steps

Step 1: Step: Inspect and replace the SFP/XFP, Command: replace optics interface GE0/0/0, Description: This command replaces the SFP/XFP on the GE0/0/0 interface. If the SFP/XFP is degraded or faulty, replacing it may resolve the issue.

Step 2: Step: Verify fiber connection, Command: display optics interface statistics GE0/0/0, Description: This command verifies the fiber connection and checks for any errors or issues. If the output indicates proper fiber connection and no errors, the issue may be resolved.

Step 3: Step: Check for Rx/Tx power loss, Command: display optics interface brief, Description: This command checks for any Rx/Tx power loss issues. If the output indicates power loss, it may be a sign of SFP/XFP degradation or improper fiber connection.

Post-Check Verification

```
Command: display optics interface statistics GE0/0/0, Expected Output:
Optics Interface Statistics for GE0/0/0
------------------------------------------------- Total Rx Errors: 0
Total Tx Errors: 0 Last Clear Time: Never , Description: This command
verifies the statistics for the GE0/0/0 interface, including total Rx
and Tx errors. The output indicates that there are no errors, which
can be a sign of proper fiber connection and SFP/XFP functionality.

Command: display optics interface brief, Expected Output: Interface Tx
Power(Rx) Rx Power(Tx) Status
------------------------------------------------- GE0/0/0 - - Present
GE0/0/1 - - Present , Description: This command verifies the optical
interface information, including Tx and Rx power levels. The output
indicates that the interface is present, which can be a sign of proper
fiber connection and SFP/XFP functionality.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP, Command: undo replace optics interface GE0/0/0, Description: This command restores the original SFP/XFP on the GE0/0/0 interface. If the replacement fails, this command can be used to revert to the original configuration.

- Step: Verify fiber connection, Command: display optics interface statistics GE0/0/0, Description: This command verifies the fiber connection and checks for any errors or issues. If the output indicates proper fiber connection and no errors, the issue may be resolved.